

Macroeconomics

Introduction and Basic Facts

(Blanchard, Macroeconomics 8th edition Ch. 1–2)

What is macroeconomics?

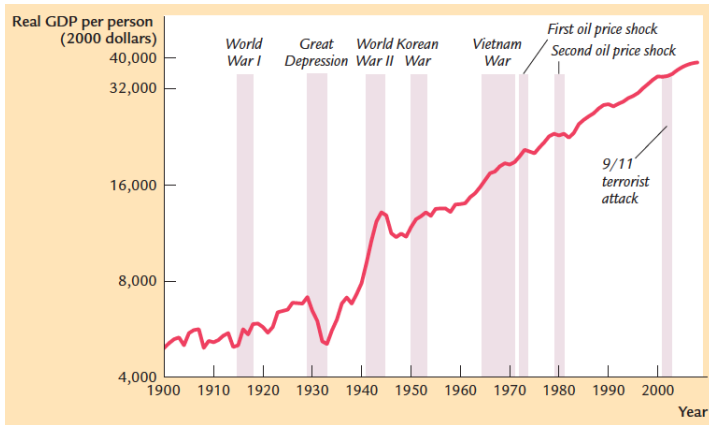
▶ **Business cycles:**

- ▶ Why do countries experience episodes of expansion and recession?
- ▶ What happens to income, unemployment, prices, etc. in the economy over business cycles?

▶ **Growth:**

- ▶ Why have some countries experienced rapid growth in incomes over the past century while others stay mired in poverty?
 - ▶ What economic policies and institutions contribute?
- ▶ State of the economy affects everyone, so macroeconomic issues play a central role in national political debates.
- ▶ Popularity of the incumbent leader often rises when the economy is doing well and falls when it is doing poorly.

Real GDP per capita, US



What is macroeconomics? (con.)

- ▶ Macroeconomic issues:
 - ▶ Why do some countries have high rates of inflation while others maintain stable prices?
 - ▶ Should China maintain a fixed exchange rate against USD?
 - ▶ Why is the US running large trade deficits?
 - ▶ Was it a good move for much of Europe to adopt a common currency?
 - ▶ How can poor nations raise their standard of living?
- ▶ Macroeconomic **policy**: How can the government reduce the frequency and severity of boom and bust cycles?

Microeconomics versus macroeconomics

- ▶ **Microeconomics:** study of how households and firms make decisions and how these decision-makers interact in marketplace.
- ▶ There is a link between the micro and the macro!
- ▶ Economy-wide events arise from the interaction of many households and firms.
- ▶ When we study the economy as a whole, we must consider the decisions of individual economic actors.
 - ▶ For example, to understand what determines total consumer spending, we must think about a family deciding how much to spend today and how much to save for the future.
 - ▶ To understand what determines total investment spending, we must think about a firm deciding whether to build a new factory.

Microeconomics versus macroeconomics (con.)

- ▶ Because aggregate variables are the sum of the variables describing many individual decisions, macroeconomic theory rests on a microeconomic foundation.
- ▶ Although microeconomic decisions underlie macroeconomic phenomena, macroeconomic models do not necessarily focus on the optimizing behavior of households and firms.
- ▶ Instead, sometimes leave that behavior in the background.

ECONOMY

New figures from the US do not point in the direction of a recession

Unemployment claims are falling in the US. Talk of recession is overblown, says market strategist.



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GEM PÅ LÆSELISTE

Pat the horse, seems to be the message to stock market investors after the United States on Thursday published the weekly figures for people seeking unemployment benefits.

Because they point to the fact that the stock markets in the US and around the world were quick enough to press the panic button earlier this week for fear of an American recession.

The inflation crisis is over. But the wild price jumps can quickly come again, according to a top Danish economist

Danmarks Nationalbank's chief economist Thomas Harr believes that for the rest of the 2020s we will have to live with greater economic turbulence, higher prices and lower growth.



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SAVE TO READING LIST



KENNETH PRAEFKE

Jyllands-Posten's business analyst

Do you remember when prices rose by 10 per cent? per year? When inflation and record interest rate hikes were all we talked about?

That is no longer the case.

But even if the inflationary crisis that has plagued the world economy for the past three years appears to be over, there is no reason to believe that we can put inflation back in the drawer with things we no longer need to worry about.

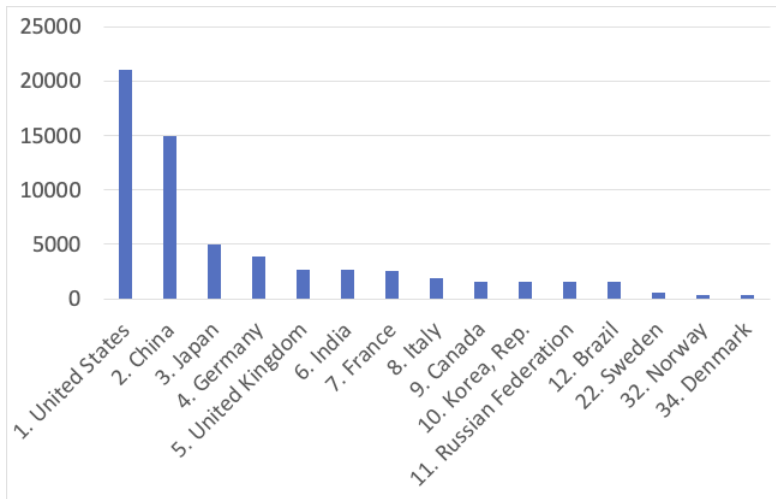
What we will cover

- ▶ Measurement: GDP, unemployment, inflation
- ▶ Keynesian model of economy and financial markets
- ▶ IS-LM model and business cycle theory
- ▶ Labor markets, inflation and the Phillips curve
- ▶ Growth and technological progress
- ▶ The role of expectations in macroeconomics
- ▶ International macroeconomics and exchange rates
- ▶ International trade
- ▶ Tutorials every week: Please inform me if any issues arise!

What is a country?

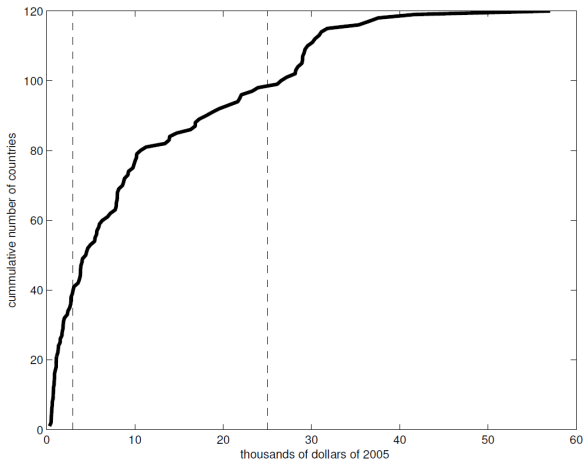
- ▶ International organizations:
 - ▶ United Nations: 193 members, includes North Korea, Cuba, excludes Taiwan
 - ▶ World Bank: 189 members, Hong Kong not a member, excludes North Korea, Cuba, Taiwan
 - ▶ World Trade Organization: 164 members, Hong Kong is member by itself; Chinese Taipei is a member (2002)
 - ▶ International Monetary Fund: 190 members, Hong Kong and Taiwan are not members
- ▶ The world economy in 2020 (WB World Development Indicators):
 - ▶ Total GDP: 85.2 trillion USD (down from 87.7 trillion in 2019)
 - ▶ Total population: 7.8 billion (up from 7.7 billion in 2019)
 - ▶ GDP per capita (PPP, constant 2017 USD): \$16,204 (down from \$16,865 in 2019)

GDP rankings (2020, billions of USD)



Distribution of income

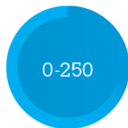
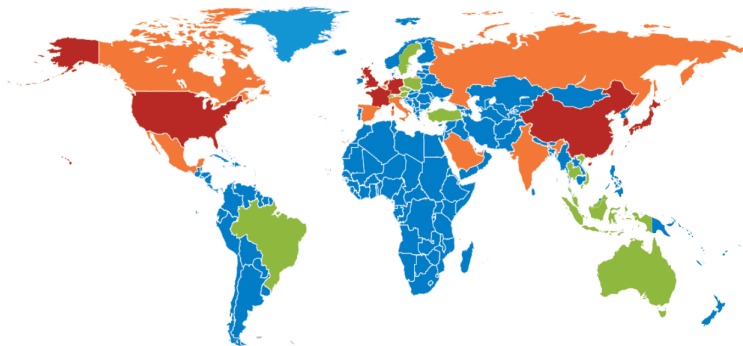
- ▶ Strong concavity of the plotted line suggests that GDP is highly unequally distributed across countries
- ▶ Few rich countries and many poor and emerging countries



Globalization over 5 centuries, $(E+M)/Y$



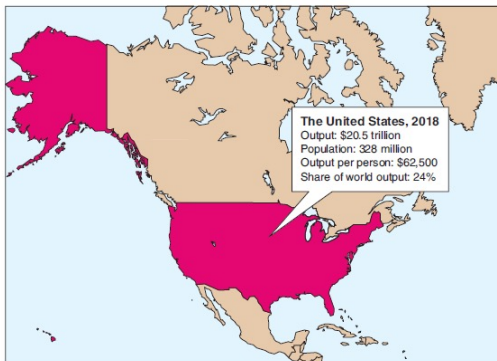
Economies by size of merchandise trade (2014)



US\$ billion

Source– WTO

The United States



- ▶ The United States is big: with an output of \$20.5 trillion in 2018, it accounted for 24% of world output.
- ▶ The U.S. standard of living is high: output per capita is \$62,500, close to the highest in the world.

The United States (con.)

- ▶ The U.S. economy in 2018 was in good shape, leaving the effects of the 2008-2009 crisis behind with one of the longest economic expansions on record.

Percent	1990–2007 (average)	2008–2009 (average)	2010–2017 (average)	2018
Output growth rate	3.0	–1.3	2.2	2.9
Unemployment rate	5.4	7.5	6.8	3.7
Inflation rate	2.3	1.3	1.6	2.3

Output growth rate: annual rate of growth of output (GDP).

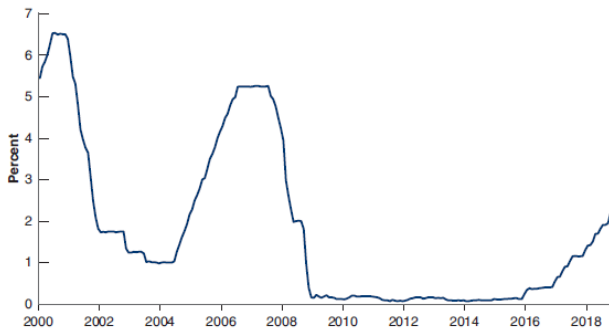
Unemployment rate: average over the year. Inflation rate: annual rate of change of the price level (GDP deflator).

Source: IMF, *World Economic Outlook*, October 2018.

Federal Funds Rate

- ▶ The federal funds rate—the interest rate the Fed controls—went from 2.5% in July 2007 to nearly 0% in December 2008.
- ▶ Why did the Federal Funds rate stop at zero?
- ▶ This constraint known as the *zero lower bound*.
- ▶ What if interest rates are negative?
- ▶ Why are low interest rates a potential issue?
 - ▶ Limit the Fed's ability to respond to further negative shocks.
 - ▶ May lead to excessive risk taking by investors to increase returns.

Federal Funds Rate



Federal Funds Rate with Market Expectations



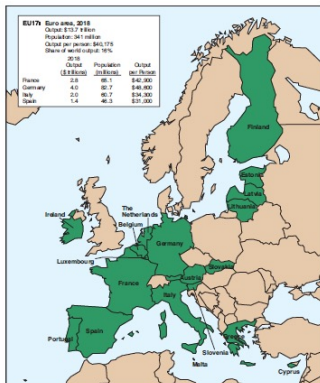
Productivity

- ▶ Productivity growth is important for a sustained increase in income per person, but since 2010, it has been only about half as it was in the 1990s.
- ▶ The slowdown in productivity growth is worrisome because the standard of living especially for the poor may not increase.

Percent change; year on year (average)	1990s	2000s	2010–2018
Private nonfarm business sector	2.2	2.8	0.9
Manufacturing	4.1	3.6	0.4

Source: FRED database. PRS85006092, MPU490063.

European Union (EU)



- ▶ 27 European countries with a common market (Brexit!)
- ▶ In 1999, the EU formed a common currency area called the Euro area, which replaced national currencies in 2002 with the euro.

Brexit

- ▶ <https://www.theguardian.com/business/2022/dec/03/swati-dhingra-bank-of-england-recession-inflation-interest-rate>
- ▶ “As the newest policymaker at the Bank of England responsible for setting interest rates, Swati Dhingra can see a problem. Britain's economy is heading for a lengthy and painful recession, while Brexit is making matters worse.
- ▶ After a year of the central bank driving up borrowing costs – the equivalent of stamping hard on the economic brakes – to tackle rampant inflation, the London School of Economics academic worries that much tougher action could intensify the crash.
- ▶ As a leading expert in international trade who wrote extensively on Brexit before joining the Bank, Dhingra says the consequences of leaving the EU – including red tape and delays at the borders, a weaker pound and stalling business investment – have worsened Britain's economic slump.”

The Euro Area (con.)

- ▶ The Euro area faces two main issues today:
 - ▶ How to reduce unemployment?
 - ▶ How to function efficiently as a common currency area?
- ▶ While the US recovered from the 2008–2009 crisis, output growth in the Euro area was negative between in both 2012 and 2013.
- ▶ In 2018, output growth was below the pre-crisis average and the unemployment rate was 8.3%.

Percent	1990–2007 (average)	2008–2009 (average)	2010–2017 (average)	2018
Output growth rate	2.1	–2.0	1.3	2.0
Unemployment rate	9.4	8.6	10.6	8.3
Inflation rate	2.1	1.5	1.0	1.5

Output growth rate: annual rate of growth of output (GDP).

Unemployment rate: average over the year. Inflation rate: annual rate of change of the price level (GDP deflator).

Source: IMF, *World Economic Outlook*, October 2018.

The Euro Area (con.)

- ▶ While the average unemployment rate for the Euro area was 8.3% in 2018, countries like Spain had an unemployment rate of 15%.
- ▶ Much of the high unemployment rate was the result of the crisis.
- ▶ Even when Spain had its lowest unemployment rate of 8%, it was nearly three times that of the Germany today.
- ▶ Some economists believe labor market rigidities with too much protection for workers are the main problem. Compare this to the Danish *flexicurity* model.
- ▶ <https://tradingeconomics.com/spain/unemployment-rate>

The Euro Area (con.)

- ▶ Supporters of the euro argue:
 - ▶ Economic advantages due to no more changes in exchange rates
 - ▶ Its contribution to the creation of a large economic power
- ▶ Others argue:
 - ▶ The drawback of a common monetary policy across euro countries
 - ▶ The loss of the exchange rate as an adjustment instrument within the euro area

EUR-DKK Exchange Rate



China

- ▶ China is in the news every day.
- ▶ Its population is more than four times that of the US, but its output at \$13.5 trillion is only about 60% of the US.
- ▶ Output per person is roughly 15% of that of the US.



China (con.)

- ▶ China has been growing very rapidly for more than three decades. But there are concerns about its growth today!
- ▶ China's rapid output growth has been driven by high accumulation of capital and technological progress.
- ▶ The slowdown after the crisis is considered to be desirable as more of output would go to consumption instead of investment.

Percent	1990–2007 (average)	2008–2009 (average)	2010–2017 (average)	2018
Output growth rate	10.2	9.4	7.9	6.6
Unemployment rate	3.3	4.3	4.1	4.0
Inflation rate	5.9	3.7	2.9	2.2

Output growth rate: annual rate of growth of output (GDP).

Unemployment rate: average over the year. Inflation rate: annual rate of change of the price level (GDP deflator).

Source: IMF, *World Economic Outlook*, October 2018.

China (con.)

<i>Average annual growth rates (%)</i>					
<i>Period</i>	<i>GDP per capita</i>	<i>Labor participation rate</i>	<i>Capital/output ratio</i>	<i>Average human capital</i>	<i>TFP</i>
1952–1978	2.97	0.11	3.45	1.55	–1.07
1978–2007	8.12	0.57	0.04	1.18	3.16
<i>Contributions to per capita GDP growth</i>					
<i>Period</i>	<i>GDP per capita</i>	<i>Labor participation rate</i>	<i>Capital/output ratio</i>	<i>Average human capital</i>	<i>TFP</i>
1952–1978	100	3.63	116.15	52.25	–72.03
1978–2007	100	7.05	0.51	14.55	77.89

Source: Authors calculations. The data on GDP per capita, GDP per worker, and labor participation rate are taken from the Penn World Table (PWT7.0). The average level of human capital is constructed using the average schooling years reported in the Barro and Lee (2010) dataset. See footnote 2 for details.

Notes: Table 1 presents a decomposition of China's per capita GDP growth into contributions from growth of labor participation rate, capital/output ratio, average human capital, and total factor productivity. "TFP" is total factor productivity. See text for details.

China (con.)

China's property crisis deepens with developer Country Garden at risk of default

Evergrande has filed for bankruptcy protection and firms covering 40% of Chinese home sales have defaulted

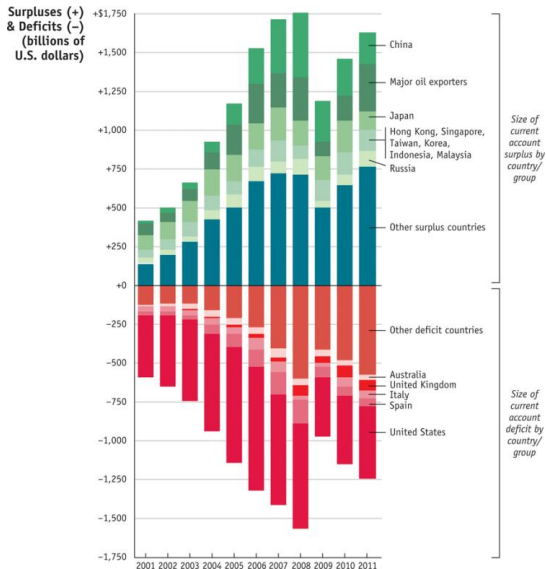


► The Country Garden logo on top of a building in Zhenjiang in China's eastern Jiangsu province.
Photograph: AFP/Getty Images

China's property crisis has deepened with two major developers facing severe financial difficulties that threaten to send shock waves through the country's economy and beyond.

- <https://www.youtube.com/watch?v=vlo0hTq4Hko>
- <https://www.theguardian.com/world/2023/aug/18/china-property-crisis-deepens-as-developer-country-garden->

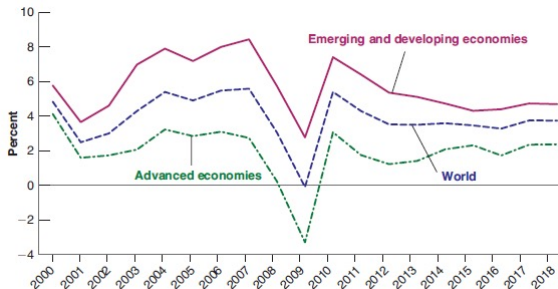
Global imbalances



Global Financial Crisis

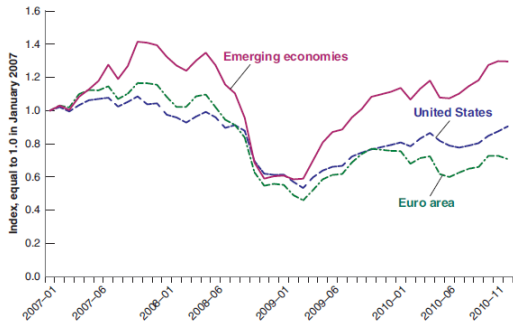
- ▶ From 2000 to 2007, the world economy had a sustained expansion.
- ▶ In 2007, U.S. housing prices started declining, leading to a major financial crisis, then major economic crisis with falling stock prices.
- ▶ The financial crisis turned into a major economic crisis with falling stock prices.
- ▶ In the third quarter of 2008, U.S. output growth turned negative and remained so in 2009.
- ▶ Through the trade and financial channels, the U.S. crisis quickly became a world crisis.

Output growth rates



Source: IMF. World Economic Outlook Database, July 2018. NGDP_RPCH.A.

Stock prices 2007-2010



Source: IMF. World Economic Outlook Database, July 2018. NGDP_RPCH.A.

Aggregate Output

- ▶ Economists studying economic activity in the 19th C. or during the Great Depression had no measure of aggregate (i.e., total) activity on which to rely.
- ▶ They had to put together bits and pieces of information, such as the shipments of iron ore, or sales at some department stores, to try to infer what was happening to the economy as a whole.
- ▶ National income and product accounts were developed at the end of World War II as measures of aggregate output.
- ▶ Measure of aggregate output: **Gross Domestic Product (GDP)**.
- ▶ How would you define aggregate output in the economy?

Two Firm Economy

Definitions of GDP

1. GDP is the value of final goods and services produced in the economy during a given period.
 - ▶ We want to count only **final** goods, not **intermediate** goods (i.e., steel).
 - ▶ If we merge the two firms in the previous example, the revenues of the new firm equal \$200.
 - ▶ This first definition gives us one way to construct GDP: by recording and adding up the production of all final goods.
 - ▶ This is roughly the way actual GDP numbers are put together.

Definitions of GDP (con.)

2. GDP is the sum of value added in the economy during a given period.
 - ▶ The value added by a firm is the value of its production minus the value of the intermediate goods used in production.
 - ▶ In two-firm example, value added equals $\$100 + \$100 = \$200$.
 - ▶ The two definitions imply that the value of final goods and services (first definition of GDP) can also be thought of as the sum of the value added by all the firms in the economy (second definition).
 - ▶ So far, we have looked at GDP from the production side.

Definitions of GDP (con.)

3. GDP is the sum of incomes in the economy during a given period.
 - ▶ Aggregate production and aggregate income are always equal.
 - ▶ Of the \$100 of value added by the steel manufacturer, \$80 goes to workers (labor income) and the remaining \$20 goes to the firm (capital income).
 - ▶ Of the \$100 of value added by the car manufacturer, \$70 goes to labor income and \$30 to capital income.
 - ▶ From the income side, valued added in the two-firm example is equal to the sum of labor income (\$150) and capital or profit income (\$50), i.e., \$200.

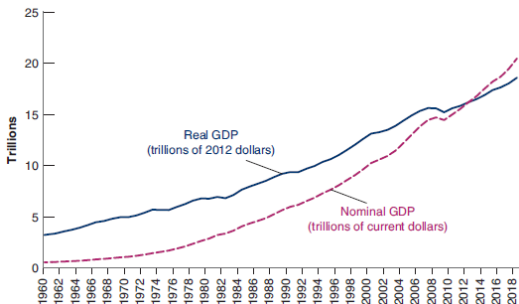
Nominal and Real GDP

- ▶ US GDP was \$20,500 billion in 2018, versus \$543 billion in 1960.
- ▶ Was US output really 38 times higher in 2018 than in 1960?
- ▶ Obviously not: Much of the increase reflected an increase in prices rather than an increase in quantities produced.
- ▶ **Nominal GDP** is the sum of the quantities of final goods produced times their current price.
- ▶ Nominal GDP increases for two reasons:
 - ▶ The production of most goods increases.
 - ▶ The price of most goods increases.
- ▶ Our goal is to measure production and its change over time.
- ▶ **Real GDP** is the sum of quantities of final goods times constant (not current) prices.

Example

US Nominal and Real GDP

- ▶ From 1960 and 2018, nominal GDP increased by a factor of 38. Real GDP increased by a factor of about 5.7.



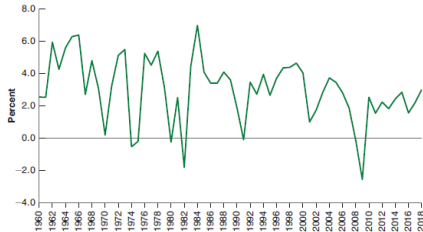
Source: FRED. Series GDPC, GDP.

Summary

- ▶ Nominal GDP is also called dollar GDP, or GDP in current dollars.
- ▶ Real GDP is also called GDP in terms of goods, GDP in constant dollars, GDP adjusted for inflation, or GDP in chained (2012) dollars, or GDP in 2012 dollars.
- ▶ GDP will refer to real GDP.
- ▶ Y_t will denote real GDP in year t .
- ▶ Nominal GDP and variables in current dollars will be denoted by a dollar sign in front of them, e.g., $\$Y_t$.

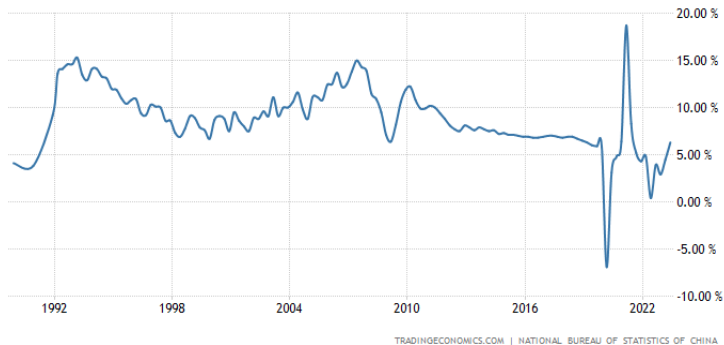
GDP: Level versus Growth Rate

- ▶ GDP growth in year t is $(Y_t - Y_{t-1})/Y_{t-1}$.
- ▶ Since 1960, the US economy has gone through a series of expansions, interrupted by short recessions.
- ▶ The 2008-2009 recession was the most severe recession in the period from 1960 to 2018.



Source: Calculated using series GDPC in Figure 2.1.

China's GDP Growth Rate



Technological Progress and Prices

- ▶ A tough problem in computing real GDP is how to deal with changes in *quality* of existing goods, e.g., computers.
- ▶ Improvements in internal speed, size of random access memory (RAM) or of the hard disk, faster access to the Internet, etc.
- ▶ *Hedonic pricing*: an approach which treats goods as providing a collection of characteristics (for computers, speed, memory, and so on), each with an implicit price
- ▶ Used by US Department of Commerce, which constructs real GDP, to estimate changes in the price of complex and fast changing goods, such as automobiles and computers.

Hedonic Pricing

- ▶ Idea: Examine market values of computers with different characteristics in a given year.
- ▶ Example: Suppose the evidence from prices of different models on the market shows that people are willing to pay 10% more for a computer with a speed of 4 GHz rather than 3 GHz.
- ▶ Suppose new computers this year have speed of 4 GHz compared to 3 GHz for new computers last year, and dollar price of new computers now is same as dollar price of new computers last year.
- ▶ (In fact, the price has declined, around 7% a year.)
- ▶ Computing the adjusted price of computers: conclude that new computers are in fact 10% cheaper than last year.

The Unemployment Rate

- ▶ **Employment** is the number of people who have a job.
- ▶ **Unemployment** is the number of people who do not have a job but are looking for one.
- ▶ The labor force is the sum of employment and unemployment.

$$\begin{array}{rcccl} L & = & N & + & U \\ \text{labor force} & = & \text{employment} & + & \text{unemployment} \end{array}$$

- ▶ The **unemployment rate** is the ratio of the number of people who are unemployed to the number of people in the labor force.

$$\begin{array}{rcccl} u & = & \frac{U}{L} \\ \text{unemployment rate} & = & \frac{\text{unemployment}}{\text{labor force}} \end{array}$$

The Unemployment Rate (con.)

- ▶ Most rich countries rely on large surveys of households to compute the unemployment rate.
- ▶ The US Current Population Survey (CPS) relies on interviews of 60,000 households every month.
- ▶ A person is unemployed if he or she does not have a job and has been looking for a job in the last four weeks.
- ▶ Those who do not have a job and are not looking for one are counted as not in the labor force: *Discouraged workers*.
- ▶ The *participation rate* is the ratio of the labor force to the total population of working age.
- ▶ Because of discouraged workers, a higher unemployment rate is typically associated with a lower participation rate.

Why Do Economists Care about Unemployment?

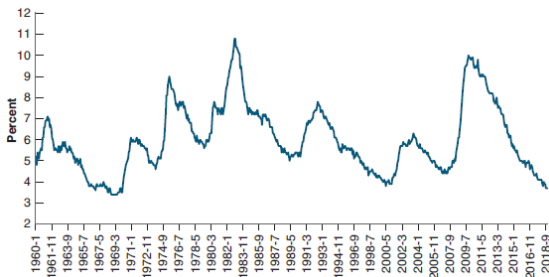
- ▶ Direct effect on the welfare of the unemployed, especially those remaining unemployed for long periods of time.
 - ▶ e.g., mean duration of unemployment, 16 weeks on average during 2000-07, increased to 40 weeks in 2011.
 - ▶ Has decreased since, but remains at a relatively high 30 weeks.
 - ▶ When the unemployment increases, not only does unemployment become both more widespread, but it also becomes more painful for those who are unemployed.
- ▶ A signal that economy is not using its human resources efficiently.

Why Do Economists Care about Unemployment?

- ▶ Very low unemployment can also be a problem as the economy runs into labor shortages.
- ▶ How low is “too low”? At the end of 2000, some economists worried that the unemployment rate (4%) was indeed too low.
- ▶ Favored lower (but positive) output growth for some time, to allow the unemployment rate to increase to a somewhat higher level.
- ▶ It turned out that they got more than they had asked for: a recession rather than a slowdown.

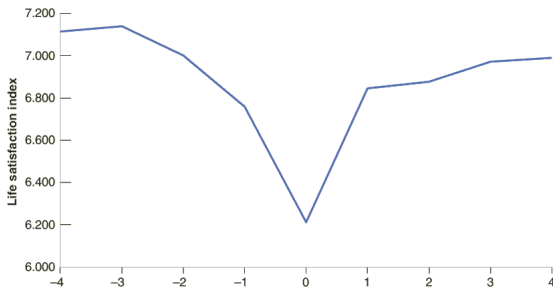
The Unemployment Rate

- ▶ Since 1960, US unemployment rate has fluctuated between 3-11%, going down during expansions and going up during recessions.
- ▶ The effect of the financial crisis is highly visible, with the unemployment rate reaching close to 10% in 2010, the highest such rate since the early 1980s.



Unemployment and Happiness

- ▶ German Socio-Economic Panel survey: followed about 11,000 households each year since 1984,
- ▶ Questions about employment status, income, and happiness.
- ▶ “How satisfied are you at present with your life as a whole?”:
 - ▶ 0 (“completely dissatisfied”) to 10 (“completely satisfied”).
- ▶ Year 0 is the year of unemployment.



Results Suggest ...

- ▶ Becoming unemployed leads to a large decrease in happiness: other studies suggest that this decrease in happiness is close to the decrease triggered by a divorce or a separation.
- ▶ Happiness declines before the actual unemployment spell: either workers know they are more likely to become unemployed, or that they like their job less and less.
- ▶ Happiness does not fully recover even four years later: unemployment may do some permanent damage, either because of the experience of unemployment itself, or because the new job is not as satisfying as the old one
- ▶ George Akerlof (Nobel Prize winner): “A person without a job loses not just his income but often the sense that he is fulfilling the duties expected of him as a human being.”

The Inflation Rate

- ▶ **Inflation** is a sustained rise in the general level of prices.
- ▶ The inflation rate is the rate at which the price level increases.
- ▶ **Deflation** is sustained decline in price level (negative inflation rate).
- ▶ The **GDP deflator** in year t (P_t) is the ratio of nominal GDP to real GDP in year t :

$$P_t = \frac{\text{Nominal GDP}_t}{\text{Real GDP}_t} = \frac{\$Y_t}{Y_t}$$

- ▶ The rate of change has a clear interpretation: the rate of inflation.

$$\pi_t = (P_t - P_{t-1})/P_{t-1}$$

The Inflation Rate (con.)

- ▶ Defining the price level as the GDP deflator implies a simple relation between nominal GP, real GDP, and the GDP deflator:

$$\$Y_t = P_t Y_t$$

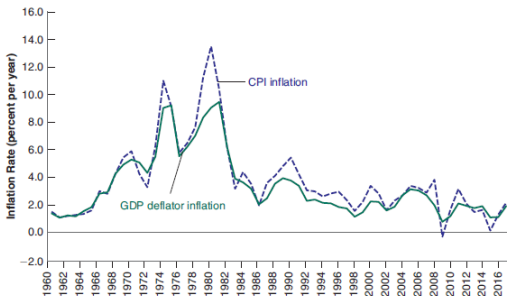
- ▶ Nominal GDP is equal to the GDP deflator times real GDP.
- ▶ The rate of growth of nominal GDP is equal to the rate of inflation plus the rate of growth of real GDP.

Consumer Price Index

- ▶ The set of goods produced in the economy is not the same as the set of goods purchased by consumers because:
- ▶ Some of the goods in GDP are sold not to consumers but to firms, to the government, or to foreigners.
- ▶ Some of the goods bought by consumers are not produced domestically but are imported from abroad.
- ▶ **Consumer Price Index** (CPI) is a measure of the cost of living.
- ▶ In US, the CPI is published monthly by Bureau of Labor Statistics (BLS), which collects price data for 211 items in 38 cities.
- ▶ The CPI gives the cost in dollars of a specific list of goods and services over time.

Comparison

- ▶ The inflation rates are largely similar.



Source: FRED: CPIAUCSL and GDPDEF

- ▶ The CPI and GDP deflator moved together most of the time.
- ▶ Exception: In 1979-80, the increase in US CPI was significantly larger than GDP deflator due to the price of imported goods increasing relative to the price of domestically produced goods.

The Inflation Rate

- ▶ Pure inflation is proportional increase in all prices and wages.
- ▶ This type of inflation causes only a minor inconvenience as relative prices are unaffected.
- ▶ Real wage (wage measured by goods rather than dollars) would be unaffected.
- ▶ Why do we care about inflation? No such thing as pure inflation.
 - ▶ Inflation affects income distribution when not all prices and wages rise proportionally.
 - ▶ Inflation leads to distortions due to uncertainty, some prices that are fixed by law or by regulation, and its interaction with taxation.
- ▶ Most economists believe the “best” rate of inflation to be a low and stable rate of inflation between 1 and 4%.

Phillips Curve

- ▶ The line is downward sloping, meaning that higher unemployment leads, on average, to a decrease in inflation, and vice versa.
- ▶ For example, if we want inflation to be 2% (e.g., current target of Fed and other central banks), the line implies that the unemployment rate has to be around 5%. But there is variance!

